

Udara Kavishka Nalawansa

Kelaniya, Western Province, Sri Lanka

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Personal Statement

Software Engineering and DevOps-focused IT undergraduate working day to day in Python, building REST APIs with FastAPI, Flask, SQLAlchemy, and PostgreSQL behind Next.js and TypeScript frontends, with Python scripting for automation. Also builds with MERN and Go and deploys with Docker, Kubernetes, and CI/CD pipelines. Designs production-ready systems, debugs and resolves live production issues, and works across frontend, backend, and infrastructure layers. Seeking a software engineering or DevOps role to deliver measurable impact in real-world systems.

Education

University of Vavuniya

2022 – Present

Bsc(Hons) in Information Technology

GPA: 3.77

St Joseph's College, Colombo 10

2005 – 2020

Professional Experience

Xaventra (Private) Limited, Colombo

Feb 2026 – Present

Software Engineer Intern

- Building and maintaining Bizadvisor, a company secretarial and business compliance platform for Sri Lankan companies, spanning a Flask admin CRM, a FastAPI backend, and a Next.js client portal and marketing site.
- Delivered features end to end, including a quotation lifecycle with PDF generation, and bulk SMS and email campaigns, using Python, Flask, SQLAlchemy, PostgreSQL, and Alembic migrations.
- Migrated the client portal to passwordless magic-link authentication with FastAPI and Next.js, integrated a WhatsApp chatbot through the Meta WhatsApp Cloud API, and automated deployments with GitHub Actions self-hosted runners monitored through Sentry.
- Designed and integrated RESTful APIs in FastAPI and PostgreSQL against a Next.js and TypeScript frontend, following OpenAPI-driven development so the generated client stays in sync with the backend contract.
- Diagnosed and resolved production issues through log analysis, database inspection, and API tracing, and supported CI/CD workflows, Alembic migrations, and incident response with root-cause analysis.

Core Competencies

Programming: Python, Java, JavaScript, TypeScript, Bash, Go

Frontend Technologies: React.js, Next.js, TypeScript, HTML, TailwindCSS, shadcn/ui, React Hook Form, Zod, TanStack Query, React Native

Backend Technologies: Node.js, Express.js, FastAPI, Flask, SQLAlchemy, Alembic, Pydantic, REST APIs, Python, Java

Databases: PostgreSQL, MySQL, MongoDB

DevOps & Cloud: Docker, Docker Compose, Podman, Kubernetes, Linux, AWS, GCP, Vercel

CI/CD & Automation: GitHub Actions (self-hosted runners), Jenkins, SonarQube, GitLab CLI, ArgoCD, Terraform, Ansible

Monitoring: Prometheus, Grafana, Sentry, ELK Stack

Tools: Git, GitHub, Claude Code (agentic coding), MCP, Postman, OpenAPI, Figma, Huly, Trello

Soft Skills: Leadership, Teamwork, Problem Solving, Adaptability, Presentation, Time Management

Projects

ParaWrite (Link)

Self Learned – 2025

- Developed a manual text paraphrasing tool to edit documents sentence by sentence with context-aware suggestions.

- Implemented features for clean text processing, user-friendly interface, and efficient workflow.

CeylonRoots (Link)

University of Vavuniya – 2025

- Built a travel agency platform using Next.js (frontend) and Spring Boot (backend).
- Designed to showcase Sri Lanka's cultural heritage while enabling travel itinerary customization, bookings, and customer support.
- Integrated responsive UI and backend APIs for seamless user interaction.

TrackNGo (MERN) (Link)

University of Vavuniya – 2025

- Developed a digital solution for public transportation in Sri Lanka with real-time bus tracking, route info, fare estimation, and lost and found management.
- Enhanced commuter experience by providing transparency, convenience, and safety through a full-stack MERN implementation.

URL Shortener (Go) (Link)

Self Learned – 2025

- Developed a high-performance URL shortening service using Go, supporting URL mapping, redirection, and persistence.
- Designed for low-latency request handling and scalability.
- Implemented clean API design and efficient data handling for fast redirections.

Kubernetes Lab Cluster Setup (Blog Post)

University of Vavuniya – 2025

- Built and configured a 4-node physical Kubernetes cluster for hands-on orchestration and research experimentation.
- Deployed workloads, managed nodes, and monitored cluster behavior under varying resource loads.
- Established the cluster as a research platform for performance testing and adaptive CPU auto-scaling experiments.

Research Experience

Undergraduate Research (BSc Hons IT)

2025 – 2026

- Completed a final year research project on **adaptive CPU auto scaling using hybrid neural networks and continual learning within Kubernetes orchestration**.
- Developed a hybrid LSTM and MLP CPU demand forecasting model with continual learning through Elastic Weight Consolidation and Experience Replay, trained on Google Cluster Traces.
- Deployed and managed a controlled Kubernetes test environment and benchmarked the predictive autoscaler against the standard Horizontal Pod Autoscaler using a Prometheus based evidence pipeline.
- Analyzed cluster behavior and resource utilization patterns to improve scaling efficiency.

Activities

Christian Society, University of Vavuniya

President | Dec 2024 – 2025

Sri Lanka University Catholic Students' Movement (SLUCSM)

Committee Member (Representative of University of Vavuniya) | 2023 – 2024

IEEE Student Branch, University of Vavuniya

Member | 2023 – Present

Leo Club of University of Vavuniya

Chief Editor | 2023 – 2024

ITCS Information Technology and Computing Society

Member | 2023 – Present

University Colours in Basketball

Awarded for outstanding performance | 2024

References

Dr. S. Kirushanth

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